# Household archetypes

**Table 1. Household profiles, mean (SD)**

| Cluster | Min. age | Max. age | Members | Education | Households | People |
|---|---|---|---|---|---|---|
| A1 | 71.6 (15.3) | 81.2 (10.7) | 1.7 (0.7) | 0.33 (0.09) | 24.2% | 17.4% |
| A2 | 17.3 (12.3) | 49.1 (10.3) | 3.2 (1.1) | 0.77 (0.10) | 32.9% | 44.3% |
| A3 | 31.0 (17.0) | 50.8 (10.7) | 2.3 (1.1) | 0.51 (0.09) | 21.8% | 20.6% |
| A4 | 56.2 (17.1) | 71.5 (9.4) | 2.0 (0.9) | 0.65 (0.12) | 21.2% | 17.7% |

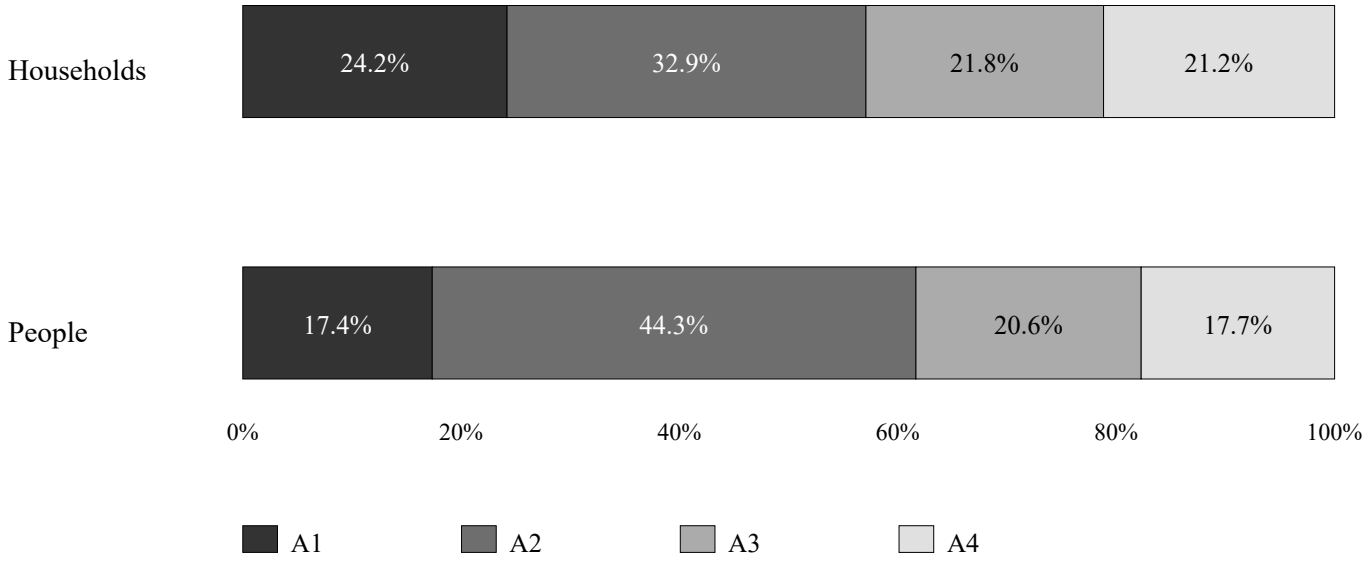



Fig. 1. Archetype shares among households and people.

*52,617 households. Shares are unweighted.*

# Dwelling area and archetype probability

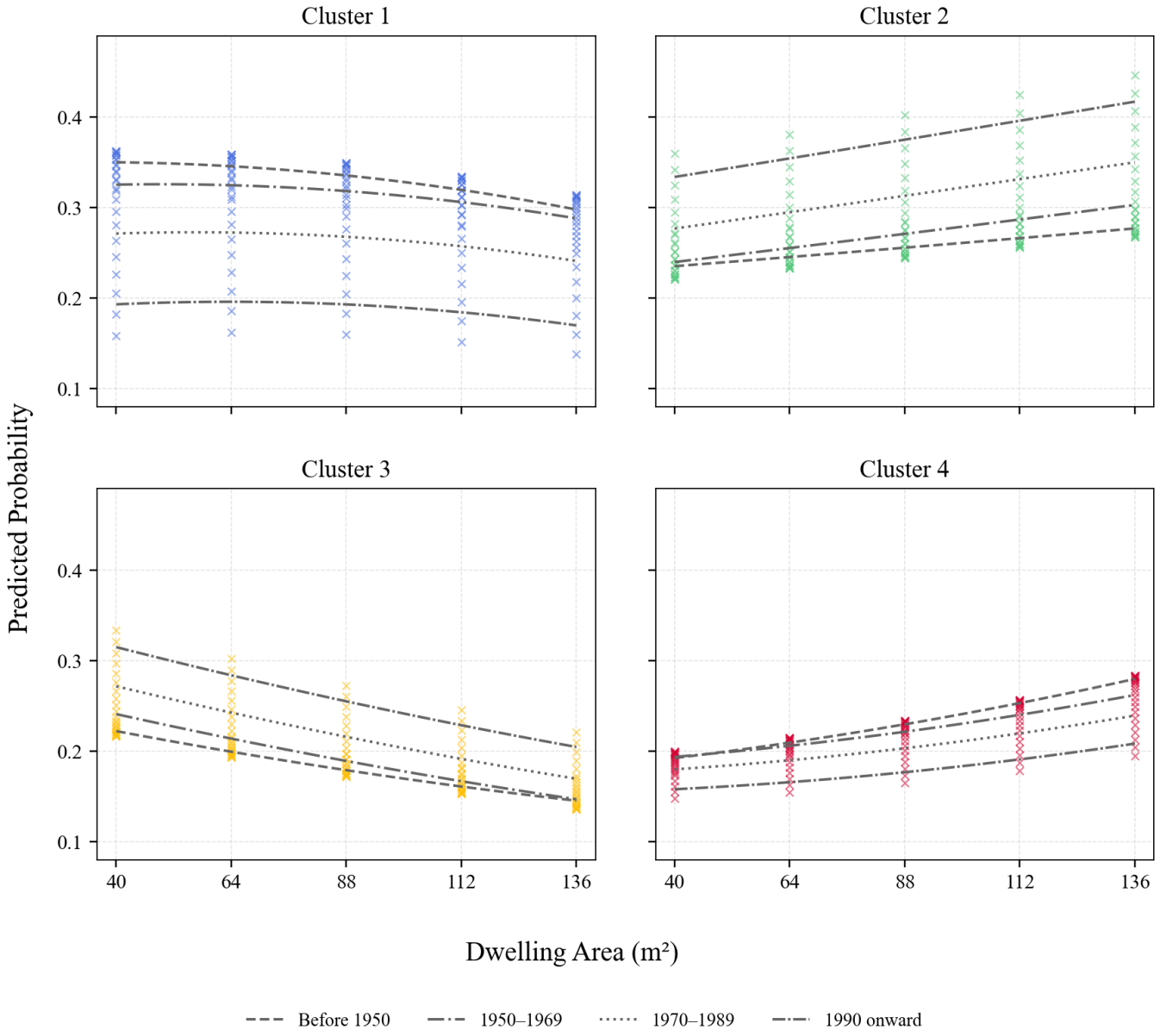



Fig. 2. Archetype probability by dwelling area and construction period.
*Shared probability scale across panels.*

# Probability model fit

### Fitted archetype probabilities

$$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix} = \begin{bmatrix} 0.05645 & 0.60268 & 0.07223 & -0.03657 & -0.30280 & -0.05015 \\ 0.38279 & -0.42585 & 0.09544 & -0.05235 & 0.24910 & 0.00172 \\ 0.43887 & -0.33479 & -0.16061 & 0.04217 & 0.15640 & 0.02055 \\ 0.12189 & 0.15796 & -0.00707 & 0.04674 & -0.10270 & 0.02789 \end{bmatrix} \times \begin{bmatrix} 1 \\ a / 100 \\ s / 100 \\ as / 10^4 \\ a^2 / 10^4 \\ s^2 / 10^4 \end{bmatrix}$$

*a = dwelling age (years); s = floor area (m²).*

*Full quadratic fit to 125 windows.*

### Table 2. Regression fit statistics

| Statistic | A1 | A2 | A3 | A4 |
|---|---|---|---|---|
| Adjusted R² | 0.901 | 0.855 | 0.946 | 0.913 |
| RMSE | 0.01847 | 0.02022 | 0.01009 | 0.01014 |

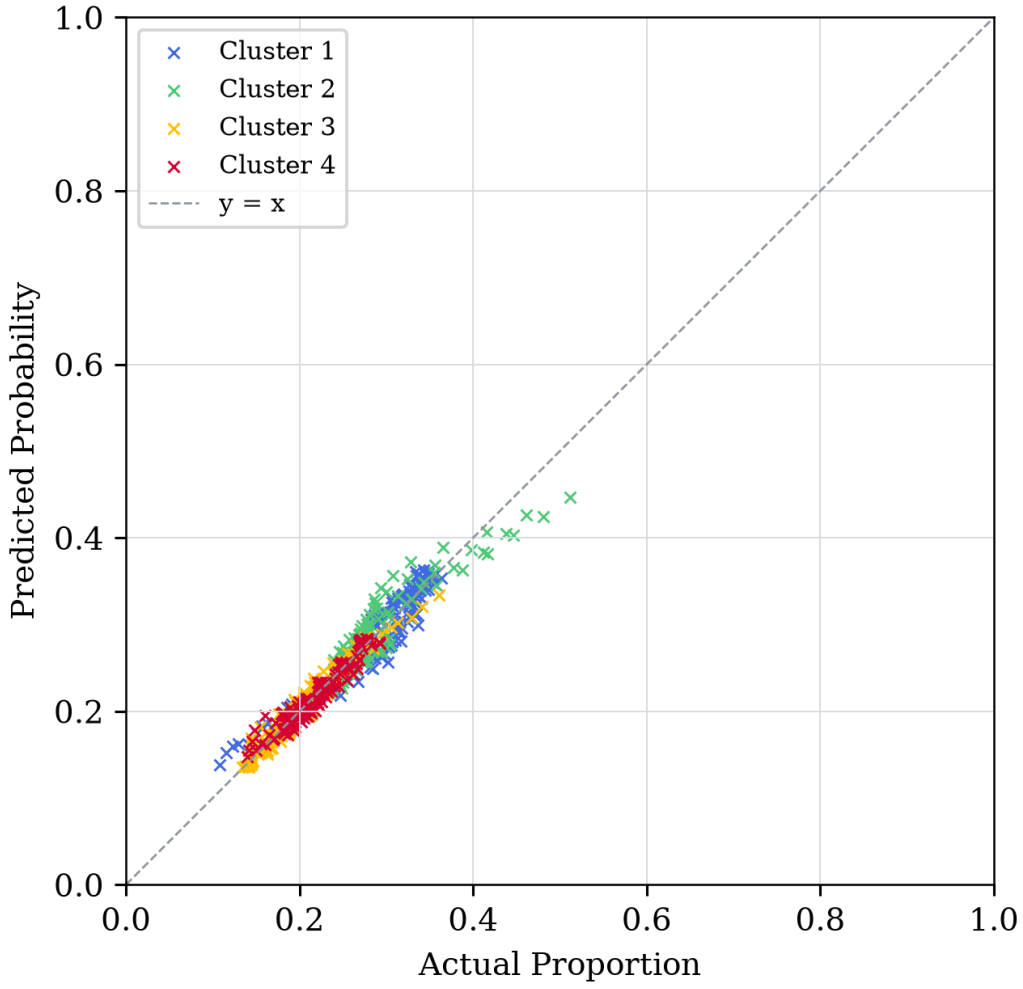



Fig. 3. Predicted probabilities and observed proportions.

# Heating groups

**Table 3. Heating system on-time (hours/day)**

| Group | Mean | SD |
|---|---|---|
| H1 | 6.42 | 3.93 |
| H2 | 7.74 | 4.43 |
| H3 | 8.96 | 5.02 |
| H4 | 10.53 | 5.43 |

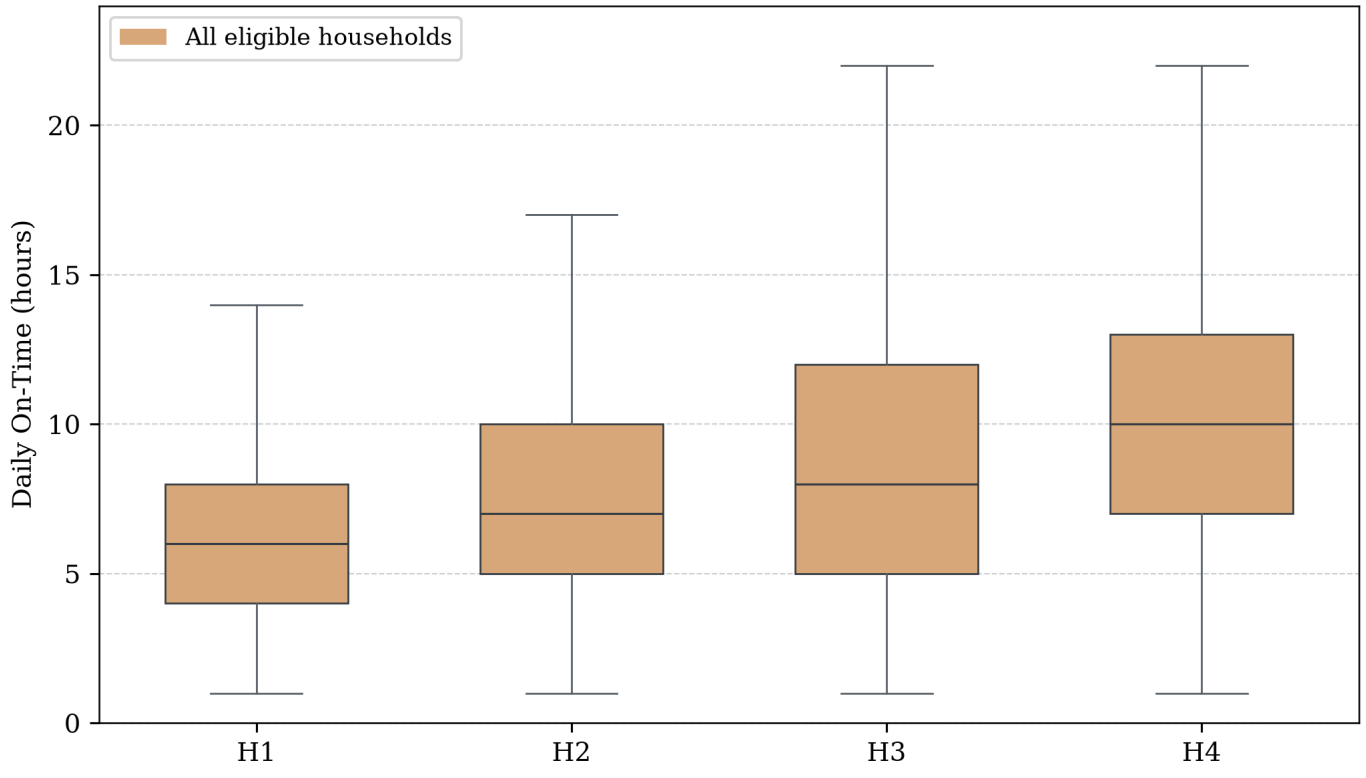



Fig. 4. Heating system on-time by group.

*Compact H1-H4 · pooled boxes · 48,690 households*

*Group labels indicate rank within each year.*

# Appliance electricity groups

**Table 4. Annual appliance consumption (kWh/year)**

| Group | Mean | SD |
|---|---|---|
| E1 | 1,664 | 560 |
| E2 | 1,856 | 649 |
| E3 | 2,088 | 739 |
| E4 | 2,275 | 796 |
| E5 | 2,691 | 872 |

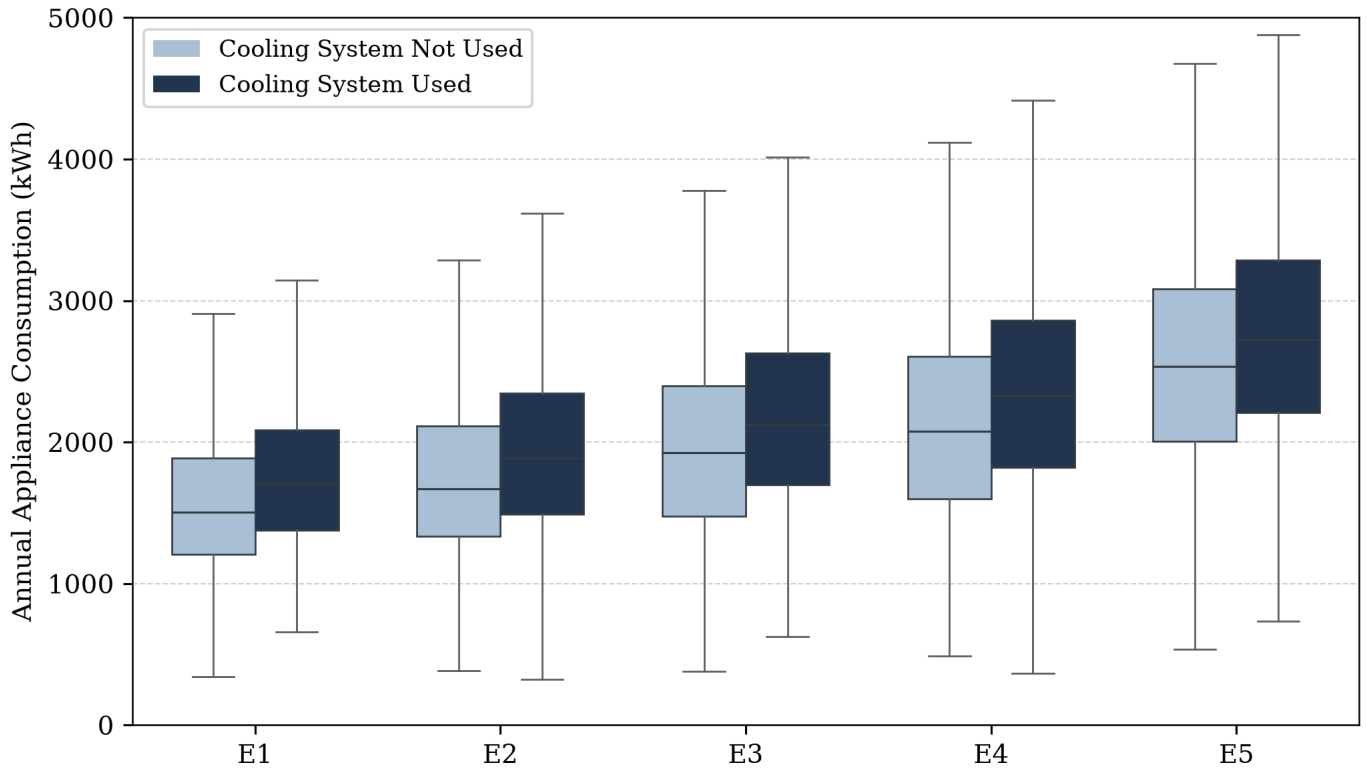



Fig. 5. Annual appliance electricity by group.

*Compact E1-E5 · 52,617 households*

*Group labels indicate rank within each year.*

# Appendix A. Group assignments

**Table A.1. Energy group by location, dwelling type, and archetype**

| Location | Dwelling type | Heating | | | | Electricity | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | A1 | A2 | A3 | A4 | A1 | A2 | A3 | A4 |
| North East | Apartment on ground, semi-basement or raised floor | H4 | H3 | H3 | H3 | E1 | E4 | E2 | E2 |
| | Apartment on intermediate floor | H4 | H3 | H3 | H4 | E1 | E4 | E2 | E2 |
| | Apartment on top floor | H4 | H4 | H3 | H4 | E1 | E4 | E2 | E2 |
| | Single-family house | H4 | H4 | H4 | H4 | E3 | E5 | E4 | E4 |
| | Multi-family house | H4 | H4 | H4 | H4 | E2 | E5 | E4 | E4 |
| North West | Apartment on ground, semi-basement or raised floor | H4 | H3 | H3 | H4 | E1 | E4 | E2 | E2 |
| | Apartment on intermediate floor | H4 | H3 | H3 | H4 | E1 | E4 | E1 | E2 |
| | Apartment on top floor | H4 | H3 | H3 | H4 | E1 | E4 | E2 | E2 |
| | Single-family house | H4 | H4 | H4 | H4 | E2 | E5 | E3 | E4 |
| | Multi-family house | H4 | H4 | H3 | H4 | E2 | E5 | E3 | E4 |
| Central | Apartment on ground, semi-basement or raised floor | H2 | H2 | H1 | H2 | E1 | E4 | E2 | E2 |
| | Apartment on intermediate floor | H1 | H1 | H1 | H2 | E1 | E4 | E2 | E2 |
| | Apartment on top floor | H2 | H2 | H1 | H2 | E1 | E4 | E2 | E2 |
| | Single-family house | H3 | H3 | H2 | H2 | E2 | E5 | E3 | E4 |
| | Multi-family house | H2 | H2 | H2 | H2 | E2 | E5 | E3 | E3 |
| South and Islands | Apartment on ground, semi-basement or raised floor | H1 | H1 | H1 | H1 | E1 | E4 | E1 | E2 |
| | Apartment on intermediate floor | H1 | H1 | H1 | H1 | E1 | E4 | E2 | E2 |
| | Apartment on top floor | H1 | H1 | H1 | H1 | E1 | E4 | E2 | E2 |
| | Single-family house | H3 | H3 | H3 | H2 | E2 | E5 | E3 | E3 |
| | Multi-family house | H2 | H2 | H2 | H2 | E2 | E5 | E3 | E3 |

Compact H1-H4 / E1-E5 · apartment classes by floor position